RESEARCH METHODOLOGY

RESEARCH DESIGN, DATA TYPES & MEASUREMENT

2 WHAT IS A RESEARCH DESIGN?

- The **blueprint** that connects RQ → data → analysis → valid answer
- Decided **before** data collection — fixing design after data is collected = bias
- Components: design type, variables & controls, data sources, instruments, analysis plan, validity safeguards
- Weak design + brilliant analysis = wrong answer with confidence.

3 MAPPING RQ TYPES TO DESIGNS

RQ type	Typical designs
Descriptive	Survey, observational study, log/telemetry analysis
Comparative	Benchmark study, controlled experiment
Relational	Correlational study (mining repos, survey + stats)
Causal	**Randomized experiment**, quasi-experiment
"Can we build X?"	Design science: build + benchmark/case study/user study

4 DESIGN GALLERY FOR CS RESEARCH

- **Survey** — questionnaires at scale (developer attitudes, tool adoption)
- **Controlled experiment** — random assignment to conditions (A/B tests, user studies)
- **Quasi-experiment** — groups exist already (Section A vs B) — weaker causal claims
- **Case study** — deep dive on one organization/system in context
- **Simulation** — when reality is too costly/dangerous (network attacks, epidemics)
- **Benchmarking** — standardized datasets + metrics (the ML default)
- **Repository mining** — GitHub/Stack Overflow data at scale
- **Action research / ethnography** — researcher embedded in practice

5 THEORY: TYPES OF DATA

Axis	Categories	CS example
Origin	**Primary** (you collect) vs **Secondary** (already exists)	Your survey vs. Kaggle dataset
Nature	**Quantitative** vs **Qualitative**	Latency numbers vs. interview transcripts
Structure	Structured / semi-structured / unstructured	SQL tables / JSON logs / video
Time	**Cross-sectional** (one snapshot) vs **Longitudinal** (over time)	One survey vs. 12-week telemetry

** Secondary data saves time but inherits someone else's collection biases — always read the dataset's documentation/datasheet.

6 MEASUREMENT SCALES (WHY THEY MATTER)

Scale	Property	Example	Allowed stats
Nominal	categories only	OS type, gender	mode, χ^2
Ordinal	ordered, unequal gaps	Likert (1–5), CGPA bands	median, Mann-Whitney
Interval	equal gaps, no true zero	temperature °C	mean, t-test
Ratio	true zero	latency ms, LOC, salary	everything, ratios

⚠️ Classic error: averaging Likert items as if interval, or running a t-test on nominal data. **Scale dictates statistics.**

7 DATA COLLECTION INSTRUMENTS

- **Questionnaires** — scalable, cheap; risk of low response & misunderstanding
- **Interviews** (structured/semi/unstructured) — rich, deep; small N, time-costly
- **Observation / think-aloud** — what users *do*, not what they *say*
- **Logs & telemetry** — objective behavior at scale; privacy implications
- **Public datasets** — UCI, Kaggle, PhysioNet/MIMIC, HuggingFace; check **license & consent provenance**
- **Instruments must be valid** (measures the right thing) **and reliable** (consistent) — Cronbach's $\alpha \geq 0.7$ for multi-item scales

8 QUESTIONNAIRE DESIGN CRASH COURSE

- One construct per question — ban **double-barreled** items ("Is the app fast and easy to use?")
- Avoid **leading** items ("Don't you agree the new UI is better?")
- Balanced **Likert scales** with labeled points; include "N/A" where honest
- Order: easy → sensitive; demographics usually last
- **Pilot test** with 5–10 people before launch — always
- Language: bilingual (Bangla/English) instruments need back-translation checks

9 THEORY: VALIDITY & RELIABILITY (VIVA FAVORITE!)

- **Internal validity** — did the IV really cause the DV change? (threats: confounds, selection bias, maturation)
- **External validity** — does it generalize beyond your sample/context?
- **Construct validity** — are you measuring what you claim? (is "LMS logins" really "engagement"?)
- **Conclusion validity** — are the statistics appropriate and powered?
- **Reliability** — would repetition give the same result?
- Good CS papers include a **"Threats to Validity"** section — yours will too.

10 CASE STUDY: CRITIQUE A FLAWED STUDY

- **Scenario:** A study claims "Dark mode improves coding speed". Method: 30 volunteers from a Discord server self-select dark or light mode, complete one task, dark group finishes 12% faster.
- **In pairs (10 min), find the flaws:**
- Self-selection (internal validity)
- Volunteers from one community (external validity)
- One task (construct validity — "coding speed"?)
- No stats test reported, N=30 split unknown (conclusion validity)
- **Fix it:** randomize assignment, multiple counterbalanced tasks, power analysis, standard environment.

11 HANDS-ON WORKSHOP (~30 MIN)

- **Part A — Questionnaire surgery (15 min):**
- Handout: a deliberately flawed 10-item questionnaire about "AI tool usage among students."
- Find ≥6 flaws (double-barreled, leading, scale abuse, missing N/A, ordering, no pilot) and rewrite 5 items correctly.
- **Part B — Group design sheet (15 min):**
- For your project RQ, complete the one-page design template: design type, data type(s), instrument, measurement scales, top-2 validity threats + mitigation.
- **Deliverable:** design sheet → submit before Week 6 with sampling decisions